- This can span longer distance at higher speeds as they have better shielding when compared to twisted pair cable.

Fiber Optical Cable

A fibre-optical cable is made of glass or plastic and transmits signals in the form of light. In fibre optical cable, light moves only in one direction. For two way communication to take place a second communication must be made between the two devices. Optical fibres use reflection to guide light through a channel. A glass or plastic core is surrounded by a cladding of less dense glass or plastic. The difference in density of the two materials must be such that a beam of light moving through the core is reflected off the cladding instead of being refracted into it.

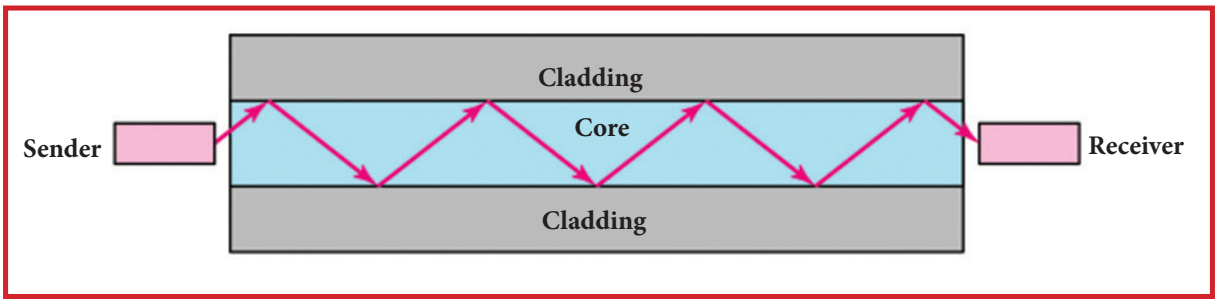


Figure 14.12 Fibre-Optical Cable

Fibre-Optic Cable Connectors

There are three types of connectors for fibre-optic cables, as shown in the figure below.

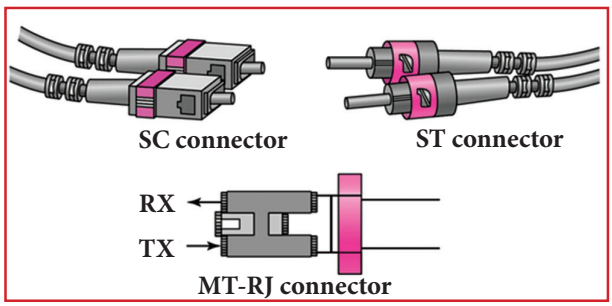


Figure 14.13 Fibre-Optical Cable Connectors

The Subscriber Channel (SC) connector is used for cable TV. It uses push/pull locking system. The Straight-Tip (ST) connector is used for connecting cable to the networking devices. The MT-RJ connector utilizes two fibres and integrates them into a single design that looks similar to an RJ45 modular connector.

Advantages

Fibre optic has several advantages over metallic cable:

- It is resistant to corrosion.
- It shows greater immunity to tapping.

Workshop-3

Go to your Computer laboratory. Identify the type of wired technology applicable there. Write a report on using the same.

14.5. Wireless Technologies

Wireless communication plays a significant role in day to day life. The term wireless refers to the communication or transmission of information over a distance without requiring wires, cables or any other electrical conductors.

The Communication is set and the information is transmitted through